

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 37-2021: A 60-Year-Old Man with Fevers, Fatigue, Arthralgias, a Mouth Ulcer, and a Rash

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PRESENTATION OF CASE

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Dr. Zoe N. Memel (Medicine): A 60-year-old man presented to this hospital because of fevers, fatigue, arthralgias, a mouth ulcer, and a rash.

The patient had been in his usual state of health until 1 month before the current presentation, when sore throat, fever, and fatigue developed while he was visiting relatives in the Southwest region of the United States. The symptoms persisted despite regular use of acetaminophen. Two weeks before the current presentation, the patient was evaluated in the emergency department of a local hospital in the Southwest, where a test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) was reportedly negative. Oral amoxicillin–clavulanic acid and doxycycline were prescribed, and the patient was discharged home.

Ten days before the current presentation, a tongue ulcer developed, along with dry scaly nodules on the face and hands and migratory arthralgias in the wrists, fingers, elbows, and shoulders. Because of worsening symptoms, the patient returned home to Boston by airplane. When he landed in Boston, the patient felt too weak to stand, and an ambulance was called to transport him to this hospital.

In the emergency department, the patient reported ongoing fevers, fatigue, arthralgias, and sore throat. He had lost 4.5 kg over a 1-month period. He had no eye redness or pain, genital ulcers, shortness of breath, cough, nausea, vomiting, diarrhea, or headache. There was no notable medical history, and the patient had not been taking any regular medications before this illness. He had completed a course of oral amoxicillin–clavulanic acid and doxycycline the day before presentation, and he had been taking acetaminophen for the past month. There were no known drug allergies.

The patient was born in China and had immigrated to the United States 20 years earlier. He lived with his wife in Boston. He had not traveled internationally for more than 10 years. The patient had previously worked in environmental maintenance but was retired. He had no history of exposure to animals or fresh water and no sick contacts. He had not been sexually active for several years. He was a smoker with a 30-pack-year history, and he drank alcohol socially; there was no

injection-drug use. His family history included colon cancer in his father, lung cancer in his mother, pancreatic cancer in his sister, and hepatocellular carcinoma in his brother.

On examination, the temperature was 37.8°C, the blood pressure 118/69 mm Hg, the heart rate 78 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The weight was 59 kg and the body-mass index (the weight in kilograms divided by the square of the height in meters) 19.2. The patient appeared unwell. There was no conjunctival injection. Oral mucous membranes were dry. There was a circular shallow ulcer on the lateral aspect of the tongue on the left side, as well as white plaque on the dorsal aspect of the tongue. There was no cervical, axillary, or inguinal lymphadenopathy. The abdomen was soft and nontender, without hepatosplenomegaly. Strength was 4/5 in both shoulders. The remainder of the neurologic examination was normal. There were several small scaly nodules with a rim of patchy erythema on the fingers, an erosion with serosanguineous crust on the left elbow, and two erythematous and mildly violaceous nodules on the forehead (Fig. 1). There were no genital lesions.

Testing of a nasopharyngeal specimen for SARS-CoV-2 RNA was negative. The ferritin level was 3089 µg per liter (reference range, 20 to 300) and the creatine kinase level 505 U per liter (reference range, 60 to 400). Urinalysis showed 2+ ketones and 1+ protein. Additional laboratory test results are shown in Table 1.

Dr. Karen Rodríguez: Chest radiography performed in the emergency department revealed bibasilar patchy opacities. Computed tomography (CT) of the chest, performed after the administration of intravenous contrast material, revealed dependent reticular opacities throughout the lower lobes of the lungs (Fig. 2). CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed a left adrenal nodule, measuring 1.2 cm in diameter.

Dr. Memel: The patient was admitted to the hospital. Intravenous fluid and oral fluconazole were administered. Biopsies of skin lesions on the left elbow and left side of the forehead were performed. During the next 5 days, the fatigue, arthralgias, skin lesions, and mouth pain continued, and daily fevers with temperatures of up

to 38.8°C were recorded. The mouth lesion did not abate with the use of fluconazole, which was being administered for presumptive oral candidiasis, and the agent was discontinued.

Dr. Samantha N. Champion: Histopathological examination of the biopsy specimen of the forehead lesion revealed vacuolar interface dermatitis with hyperkeratosis, dyskeratosis, pigment incontinence, and sparse perivascular chronic inflammation. Similar findings were observed on histopathological examination of the biopsy specimen of the elbow lesion.

Dr. Memel: Cough developed, and pain in the left shoulder increased. On the fifth hospital day, the creatine kinase level was 906 U per liter. Other laboratory test results are shown in Table 1. The next day, imaging studies of the left shoulder were obtained.

Dr. Rodríguez: Magnetic resonance imaging (MRI) of the left shoulder, performed on hospital day 6, revealed patchy mild intramuscular edema that was most pronounced in the trapezius and deltoid, a finding compatible with myositis (Fig. 2). There was no focal or organized fluid collection.

Dr. Memel: On hospital day 10, biopsy of the left deltoid was performed with the patient under general anesthesia. In the 24 hours after the muscle biopsy, hypoxemia developed, and the patient received 3 liters of supplemental oxygen through a nasal cannula to maintain an oxygen saturation of more than 93%. The next day, the patient received 8 liters of supplemental oxygen through a nasal cannula. Additional laboratory test results are shown in Table 1. Imaging studies of the chest were obtained.

Dr. Rodríguez: Repeat chest radiography revealed progressive patchy opacities at the lung bases. Repeat CT of the chest revealed progressive peribronchovascular and peripheral ground-glass and consolidative opacities predominantly involving the lower lobes, a finding consistent with organizing pneumonia (Fig. 2). In addition, new small left and trace right pleural effusions were present.

Dr. Memel: A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Zachary S. Wallace: This previously healthy 60-year-old man presented with a rapidly evolving multisystem disease. Several aspects of his



Figure 1. Clinical Photographs.

Panels A and B show several small scaly nodules with a rim of patchy erythema on the fingers. Panel C shows areas of hyperkeratosis on the lateral aspect of the fingers. Panel D shows an erosion with serosanguineous crust on the left elbow. Panel E shows two dull erythematous and mildly violaceous nodules on the forehead. Panel F shows a circular shallow ulcer on the lateral aspect of the tongue on the left side, as well as white plaque on the dorsal aspect of the tongue.

presentation — including fever, weight loss, pharyngitis, and inflammation (with an elevated erythrocyte sedimentation rate, C-reactive protein level, and ferritin level) — are common findings of infection, cancer, rheumatic disease, and other conditions. These findings are nonspecific and do not help to narrow the differential diagnosis. Migratory arthralgias are also nonspecific, especially in the absence of joint tenderness, swelling, and warmth, features that might indicate synovitis. The low titer of antinuclear antibodies (ANA) in a speckled pattern is a nonspecific test result that can be observed in healthy persons, as well as in patients with rheumatic disease, infection, cancer, or single-organ autoimmune diseases such as Hashimoto's thyroiditis.

Four features of this patient's presentation have specificity for certain diseases and thus may help to narrow the differential diagnosis. First, laboratory testing revealed the presence of anti-Ro antibodies. Second, the patient had features of myositis, including the inability to stand on his own, shoulder weakness on examination, evidence of muscle injury on laboratory testing (i.e., elevated aspartate aminotransferase and creatine kinase levels), and muscle edema on MRI. Third, he had several mucocutaneous manifestations of disease, including a tongue ulcer, as well as ulcerations and nodules on the face and hands. The white tongue plaque may also be a feature of his underlying disease. Although oral candidiasis was initially considered, the plaque did not abate with fluconazole therapy, and this lack of response would be atypical in a presumably immunocompetent host. Fourth, the patient had a rapidly evolving lung disease characterized by progressive hypoxemia.

Given these four features in particular, I would consider a rheumatic disease in this patient. However, when evaluating a patient for rheumatic disease, I also include infection, cancer, and drug reactions in my initial differential diagnosis. These conditions often mimic rheumatic disease, can be associated with the development of rheumatic disease (e.g., hepatitis B virus–associated polyarteritis nodosa, cancer-associated dermatomyositis, and drug-induced lupus erythematosus), and may be exacerbated or made more difficult to diagnose after glucocorticoids or other immunosuppressive medications are administered.

INFECTION

In this patient, the rapid onset of disease, history of travel to the Southwest, fever, and cutaneous nodules could be suggestive of an infection such as coccidioidomycosis. However, an extensive evaluation for infectious diseases was negative.

CANCER

Lymphoma, leukemia, plasma-cell dyscrasias, and solid-organ cancers should be considered in this case because of the fever, weight loss, elevated alkaline phosphatase level, adrenal nodule, smoking history, and extensive family history of cancer. However, the patient's blood counts were normal, he did not have lymphadenopathy, and key features of his presentation, especially the myositis and the pattern of lung disease, are not typical of cancer.

DRUG REACTIONS

Although the patient had been exposed to several antimicrobial agents, his mucocutaneous lesions are not typical of a drug reaction. His illness began before any drug exposure, and lung disease and myositis would not be expected features of a drug reaction.

RHEUMATIC DISEASES ASSOCIATED WITH ANTI-RO ANTIBODIES

Because infection, cancer, and drug reactions are unlikely diagnoses in this case, I will next consider how the development of anti-Ro antibodies, myositis, mucocutaneous lesions, and rapidly progressive lung disease may best fit with a diagnosis of rheumatic disease. The most common rheumatic diseases associated with anti-Ro antibodies are Sjögren's syndrome and systemic lupus erythematosus. Less common diseases associated with anti-Ro antibodies are inflammatory myositis, mixed connective-tissue disease, and primary biliary cholangitis.

Could Sjögren's syndrome or lupus explain this patient's presentation? Sjögren's syndrome typically affects women in their sixth decade and is characterized by an insidious onset of symptoms, including dry eyes and mouth.¹ Extraglandular features may include nonspecific constitutional symptoms, pain, and neuropathy. The myositis, rapidly progressive lung disease, and mucocutaneous lesions observed in this patient would be atypical of Sjögren's

Table 1. Laboratory Data.*

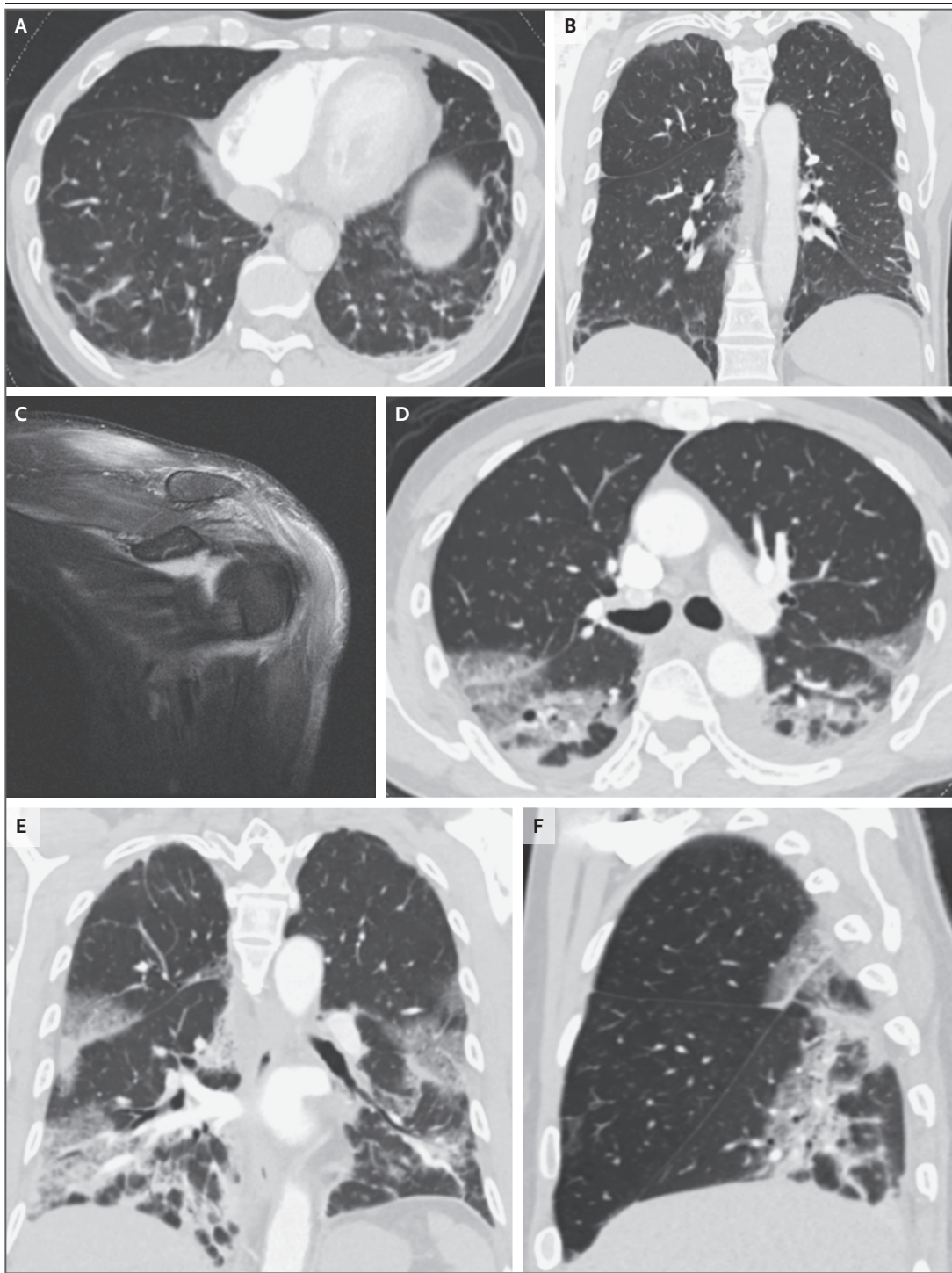
Variable	Reference Range, Adults†	On Admission	Hospital Days 1–5	Hospital Day 11
Blood				
Hemoglobin (g/dl)	12.0–16.0	13.0	—	10.5
Hematocrit (%)	41.0–53.0	39.1	—	32.4
Mean corpuscular volume (fl)	80.0–100.0	88.3	—	88.7
White-cell count (per μ l)	4500–11,000	4900	—	6370
Differential count (per μ l)				
Neutrophils	1800–7700	3850	—	5180
Immature granulocytes	0–100	70	—	100
Lymphocytes	1000–4800	420	—	520
Eosinophils	0–900	70	—	0
Basophils	0–300	10	—	10
Monocytes	200–1200	480	—	560
Platelet count (per μ l)	150,000–450,000	169,000	—	243,000
Creatinine (mg/dl)	0.60–1.50	0.69	—	0.54
Urea nitrogen (mg/dl)	8–25	16	—	16
Aspartate aminotransferase (U/liter)	9–32	165	—	204
Alanine aminotransferase (U/liter)	7–33	165	—	112
Alkaline phosphatase (U/liter)	30–100	125	—	232
Total bilirubin (mg/dl)	0.0–1.0	0.5	—	0.4
Albumin (g/dl)	3.3–5.0	3.3	—	2.6
Globulin (g/dl)	1.9–4.1	3.1	—	3.1
Ferritin (μ g/liter)	20–300	3089	—	—
Creatine kinase (U/liter)	60–400	505	906	3220
High-sensitivity troponin T (ng/liter)	0–14	15	—	—
Lactate dehydrogenase (U/liter)	110–210	386	—	—
C-reactive protein (mg/liter)	<8.0	19.5	—	—
Erythrocyte sedimentation rate (mm/hr)	0–20	29	—	—
Thyrotropin (μ IU/ml)	0.40–5.00	0.99	—	—
IgG (mg/dl)	614–1295	—	980	—
IgA (mg/dl)	69–309	—	220	—
IgM (mg/dl)	53–334	—	50	—
Serum protein electrophoresis	—	—	Normal	—
C3 (mg/dl)	81–157	—	91	—
C4 (mg/dl)	12–39	—	40	—
HIV type 1 and 2 antibody and antigen	Nonreactive	—	Nonreactive	—
Treponemal antibody	Nonreactive	—	Nonreactive	—
Brucella IgG and IgM	Negative	—	Negative	—
<i>Coxiella burnetii</i> phase I and II IgG and IgM	Negative	—	Negative	—
Coccidioides antibody	Negative	—	Negative	—
Cytomegalovirus IgG and DNA, PCR assay	Negative	—	Negative	—
Epstein–Barr virus DNA, PCR assay	Negative	—	Negative	—

Table 1. (Continued.)

Variable	Reference Range, Adults†	On Admission	Hospital Days 1–5	Hospital Day 11
Hepatitis B virus				
Surface antibody	Negative	Negative	—	—
Surface antigen	Negative	Negative	—	—
Core antibody	Negative	Positive	—	—
DNA	Not detected	Not detected	—	—
Hepatitis C virus antibody	Negative	Negative	—	—
Parvovirus B19 DNA	Negative	—	Negative	—
Tuberculosis antigen, IGRA	Negative	—	Negative	—
Blood cultures	Negative	Negative	Negative	—
Babesia, thick and thin smear	Negative	—	Negative	—
Lyme disease antibody	Negative	—	Negative	—
SARS-CoV-2 IgG and IgM	Negative	—	Negative	—
Cryoglobulins	None present	—	None present	—
Antinuclear antibody				
Qualitative	Negative	—	Positive	—
Quantitative	—	—	1:40–1:160 (speckled pattern)	—
Antineutrophil cytoplasmic antibody	Negative	—	Negative	—
Anti–double-stranded DNA antibody	Negative	—	Negative	—
Anti-SSA (Ro) antibody				
Qualitative	Negative	—	Positive	—
Quantitative (OD units)	0.0–19.0	—	104.04	—
Anti-SSB (La) antibody	Negative	—	Negative	—
Anti-Smith antibody	Negative	—	Negative	—
Anti–U1-ribonucleoprotein antibody	Negative	—	Negative	—
Anti–Scl-70 antibody	Negative	—	Negative	—
Anti–Jo-1 antibody	Negative	—	Negative	—
Throat swab				
<i>Neisseria gonorrhoeae</i> DNA	Negative	—	Negative	—
<i>Chlamydia trachomatis</i> DNA	Negative	—	Negative	—
Herpes simplex virus type 1 DNA	Negative	—	Negative	—
Herpes simplex virus type 2 DNA	Negative	—	Negative	—
Swab culture	Negative	—	Negative	—
Urine				
Histoplasma antigen	Negative	—	Negative	—
Blastomyces antigen	Negative	—	Negative	—
Urine culture	Negative	—	Negative	—

* To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. HIV denotes human immunodeficiency virus, IGRA interferon- γ release assay, OD optical density, PCR polymerase chain reaction, and SARS-CoV-2 severe acute respiratory syndrome coronavirus 2.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.



syndrome, especially in the absence of dry eyes and mouth.

Like Sjögren's syndrome, lupus often has an insidious onset of symptoms and is more common in women than in men, although the dis-

ease typically develops during the childbearing years.² Mucocutaneous manifestations are common, but the typical lesions observed in patients with lupus (e.g., acute or subacute cutaneous lupus or discoid lupus) are distinct, in terms of

Figure 2 (facing page). Imaging Studies.

CT was performed on hospital day 1. An axial image through the lower lungs (Panel A) shows dependent reticular and ground-glass opacities. A coronal image through the mid-lungs (Panel B) shows peribulbar thickening and patchy ground-glass opacities and septal thickening without honeycombing or fibrosis. MRI was performed on hospital day 6. A T2-weighted oblique image of the left shoulder (Panel C) shows patchy mild intramuscular edema that is most pronounced in the trapezius and deltoid. Repeat CT was performed on hospital day 11. An axial image through the mid-lungs and lower lungs (Panel D) shows progressive peribronchovascular and peripheral ground-glass and consolidative opacities, a finding compatible with organizing pneumonia, as well as small left and trace right pleural effusions. Coronal and sagittal images (Panels E and F, respectively) show peribulbar and septal thickening and consolidation.

distribution and features, from the nodules and ulcerations described in this patient; he had a tongue ulcer, as opposed to the buccal mucosal ulcers often seen in patients with lupus. In addition, other common features of lupus — including cytopenias, the presence of autoantibodies (e.g., antibodies directed against Smith antigen and double-stranded DNA), and hypocomplementemia — were absent in this case. Therefore, although lupus and Sjögren's syndrome are commonly associated with anti-Ro antibodies, neither disease is a good fit with this patient's presentation.

RHEUMATIC DISEASES ASSOCIATED WITH MYOSITIS

In one approach to the evaluation of a patient with myositis, the first step is to characterize the distribution of muscle involvement as symmetric or asymmetric. Pyomyositis and diabetic muscle infarction tend to have an asymmetric distribution, whereas rheumatic diseases tend to be associated with symmetric myositis. The exception is inclusion-body myositis, which tends to be asymmetric.

In patients with symmetric myositis, such as this patient, the next step in the evaluation is to characterize the distribution as primarily proximal or distal. With rheumatic diseases, except for inclusion-body myositis, proximal rather than distal muscles tend to be involved. Endocrinopathies, such as hypothyroidism, can also cause symmetric proximal myopathy, but this patient had a normal thyrotropin level.

Overall, the symmetric proximal myositis in this patient is consistent with idiopathic inflammatory myopathy, which has been associated with anti-Ro antibodies.³ The classic categories of idiopathic inflammatory myopathy are inclusion-body myositis, polymyositis, overlap syndromes (e.g., Sjögren's syndrome and lupus), necrotizing myositis (e.g., statin-induced myositis), antisynthetase syndrome, and dermatomyositis.

Inclusion-body myositis is an unlikely diagnosis in this case because of the distribution of muscle involvement. Polymyositis is unlikely because of the mucocutaneous manifestations. Overlap syndromes are unlikely because of the absence of characteristic features of conditions such as Sjögren's syndrome, lupus, systemic sclerosis, and others. The patient had not been treated with statins and did not have features of necrotizing myositis on MRI. Patients with antisynthetase syndrome typically have myositis along with other classic features, including Raynaud's phenomenon, mechanic's hands (hyperkeratosis on the sides of the digits),⁴ interstitial lung disease, arthritis, and fever.⁵ This patient did not have Raynaud's phenomenon or inflammatory arthritis, and his ulcerative and nodular skin lesions would not be expected features of antisynthetase syndrome; the rapid progression of his lung disease would also be atypical.

DERMATOMYOSITIS

Could this patient have dermatomyositis? Typical cutaneous manifestations of this disease include Gottron's papules (erythematous papules on the metacarpophalangeal and proximal interphalangeal joints and other extensor surfaces), heliotrope rash (an erythematous-to-purplish rash that often involves the eyelids and surrounding area), V sign (also known as shawl sign, an erythematous rash on the front and back of the torso), and holster sign (an erythematous rash that involves the lateral hip area). This patient did not have these typical cutaneous manifestations of dermatomyositis. However, dermatomyositis can be associated with interstitial lung disease, which was present in this patient. Although dermatomyositis remains a possible diagnosis in this case, I will consider other rheumatic diseases that could explain his presentation.

OTHER RHEUMATIC DISEASES ASSOCIATED WITH MYOSITIS

Adult-onset Still's disease, sarcoidosis, and vasculitis can all be associated with myositis. However, myositis, lung disease, and the presence of anti-Ro antibodies are not typical manifestations of adult-onset Still's disease, and the characteristic evanescent rash was absent in this patient.⁶ Sarcoidosis typically affects younger patients and classically causes hilar lymphadenopathy, which was absent in this patient.⁷ In addition, this patient's skin lesions and the presence of anti-Ro antibodies are not typical of sarcoidosis.

Vasculitis is a broad category of diseases associated with inflammation of the walls of small, medium, and large vessels.⁸ Polyarteritis nodosa is a consideration in this patient because he had a history of hepatitis B, on the basis of the positive test for hepatitis B virus core antibodies. Patients with polyarteritis nodosa often have myalgias, but myositis is less common, and the lung disease observed in this patient would be an unusual feature of polyarteritis nodosa.

Immune complex vasculitis, such as IgA or cryoglobulinemic vasculitis, is an unlikely diagnosis in this case, given the normal C3 and C4 levels. Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis is also unlikely, given that an ANCA test was negative. Some patients with limited forms of ANCA-associated vasculitis (e.g., disease limited to the head and neck) may have a negative ANCA test, but this patient did not have other, more typical features of the disease.⁹ Finally, Behçet's disease can be associated with oral ulcers, but lung disease and the presence of anti-Ro antibodies are not consistent with this diagnosis.

Can the results of the skin biopsies help to guide us toward the most likely diagnosis? The biopsy specimens did not show features suggestive of small- or medium-vessel vasculitis. There were no granulomas, which would be suggestive of sarcoidosis. There was evidence of interface dermatitis, which is typically associated with either lupus or dermatomyositis. Because lupus is unlikely to explain this patient's presentation, I will revisit dermatomyositis as a possible diagnosis.

RECONSIDERING DERMATOMYOSITIS

Another approach to the evaluation of this patient with idiopathic inflammatory myopathy is to consider the phenotypes of myositis associ-

ated with various myositis-specific antibodies.¹⁰ Some myositis-specific antibodies (anti-transcriptional intermediary factor 1 γ [anti-TIF-1 γ] antibodies, anti-Mi-2 antibodies, anti-small ubiquitin-like modifier-activating enzyme [anti-SAE] antibodies, and anti-nuclear matrix protein 2 [anti-NXP-2] antibodies) are associated with dermatomyositis with typical cutaneous manifestations, but these manifestations were not present in this patient. Another type (anti-histidyl-transfer RNA synthetase [anti-Jo-1] antibodies) is associated with antisynthetase syndrome, but the patient did not have the typical features of this condition, such as inflammatory arthritis or Raynaud's phenomenon. Additional myositis-specific antibodies (anti-3-hydroxy-3-methylglutaryl-coenzyme A reductase [anti-HMG-CoA reductase] antibodies and anti-signal recognition particle [anti-SRP] antibodies) are associated with necrotizing myositis, which was also absent in this case.

Anti-melanoma differentiation-associated protein 5 (anti-MDA-5) antibodies are associated with dermatomyositis with unique cutaneous features, especially skin ulcerations and nodular lesions, along with rapidly progressive interstitial lung disease and subtle myopathy; indeed, many cases of anti-MDA-5 dermatomyositis are amyopathic. Cutaneous ulcerations overlying the metacarpophalangeal and interphalangeal joints, elbows, and nail folds are seen in up to 85% of patients with anti-MDA-5 dermatomyositis, often with lateral digital hyperkeratosis, which was observed in this patient.^{11,12} Approximately 50% of cases are associated with oral ulcers, often on the tongue. The arthralgias, fever, presence of anti-Ro antibodies, abnormal results on liver-function tests, and elevated ferritin level seen in this patient are common in patients with anti-MDA-5 dermatomyositis.

I suspect that the most likely diagnosis in this patient is anti-MDA-5 dermatomyositis. To establish this diagnosis, I would review the results of the muscle biopsy and test for antibodies directed against myositis-associated autoantigens, including MDA-5.

DR. ZACHARY S. WALLACE'S
DIAGNOSIS

Anti-melanoma differentiation-associated protein 5 (anti-MDA-5) dermatomyositis.

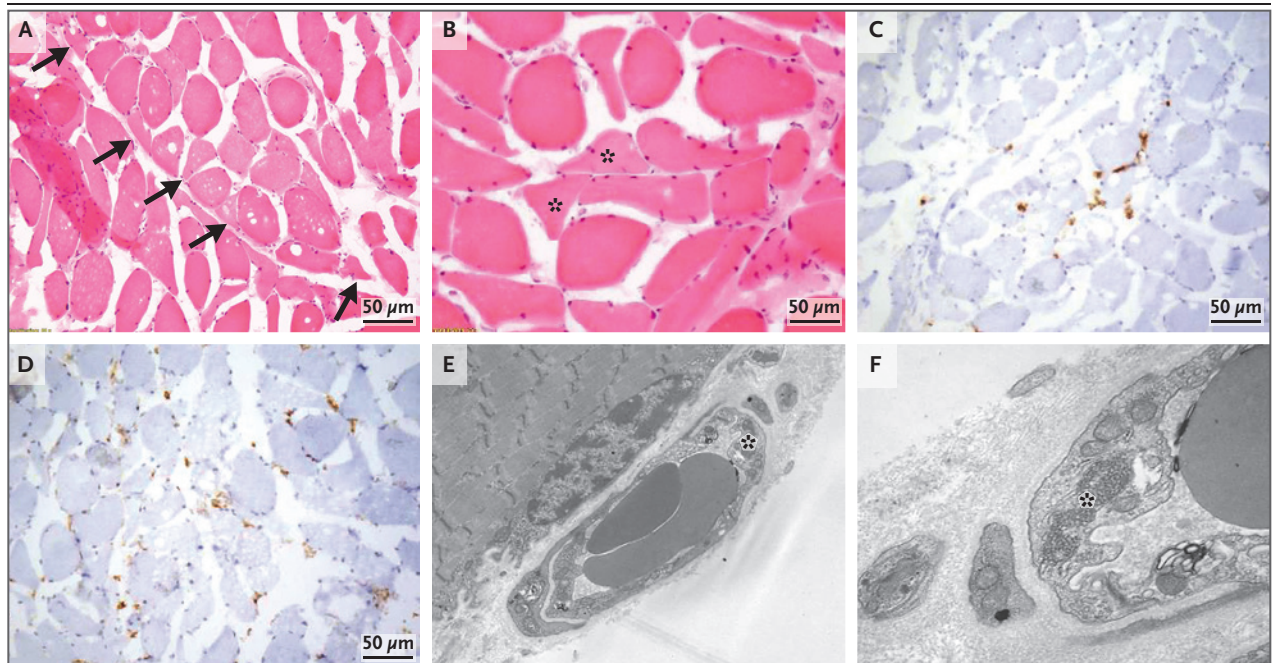


Figure 3. Biopsy Specimen from the Left Deltoid.

Hematoxylin and eosin stains of sections of frozen tissue show skeletal muscle with perifascicular atrophy (Panel A, arrows) and variation in the size and shape of muscle fibers, with scattered angulated and atrophic fibers (Panel B, asterisks). Immunohistochemical stains of sections of formalin-fixed tissue show very few signs of chronic inflammation, including sparse CD3-positive T cells (Panel C) and scattered CD68-positive perivascular macrophages (Panel D). Electron microscopic images show endothelial cells with tubuloreticular inclusions (Panels E and F, asterisks).

PATHOLOGICAL DISCUSSION

Dr. Champion: The biopsy specimen from the left deltoid showed variation in the size and shape of muscle fibers, with scattered angulated and atrophic fibers. The atrophic fibers had a preferential distribution in the perifascicular areas (Fig. 3). There was no evidence of inflammation, vasculitis, or necrotic or regenerating fibers. Immunohistochemical staining of the biopsy specimen for inflammatory markers showed only sparse CD3-positive T cells and scattered CD68-positive perivascular macrophages. Immunohistochemical staining for complement (C5b-9) was negative for deposits in capillary walls. Electron microscopic examination revealed multiple endothelial cells that contained tubuloreticular inclusions.

Despite the absence of inflammation and active injury to myocytes, the presence of perifascicular atrophy and endothelial inclusions in the muscle, in combination with the dermatopathological findings, is consistent with the diagnosis

of dermatomyositis.¹³ The absence of active disease could be due to sampling because inflammation can be patchy, or the disease could be in a quiescent phase.

Dr. Donald B. Bloch: An enzyme-linked immunosorbent assay (ELISA) was used to detect myositis-associated autoantibodies. The patient's serum contained antibodies directed against MDA-5 and Ro-52 (Fig. 4). Both of these proteins have important roles in innate cellular immunity. MDA-5 is a cytoplasmic protein that is able to sense double-stranded viral RNA and subsequently contributes to antiviral immunity by up-regulating interferon-stimulated genes.¹⁴ Ro-52 is an E3 ubiquitin ligase that binds to viruses that have been coated with antibodies and causes degradation of capsid proteins. Loss of the viral capsid proteins leads to the premature release of viral nucleic acids, which may then be sensed by proteins such as MDA-5.¹⁵ The potential roles of MDA-5 and Ro-52 in the pathogenesis of dermatomyositis are unknown.

A recent study showed the prevalence of false

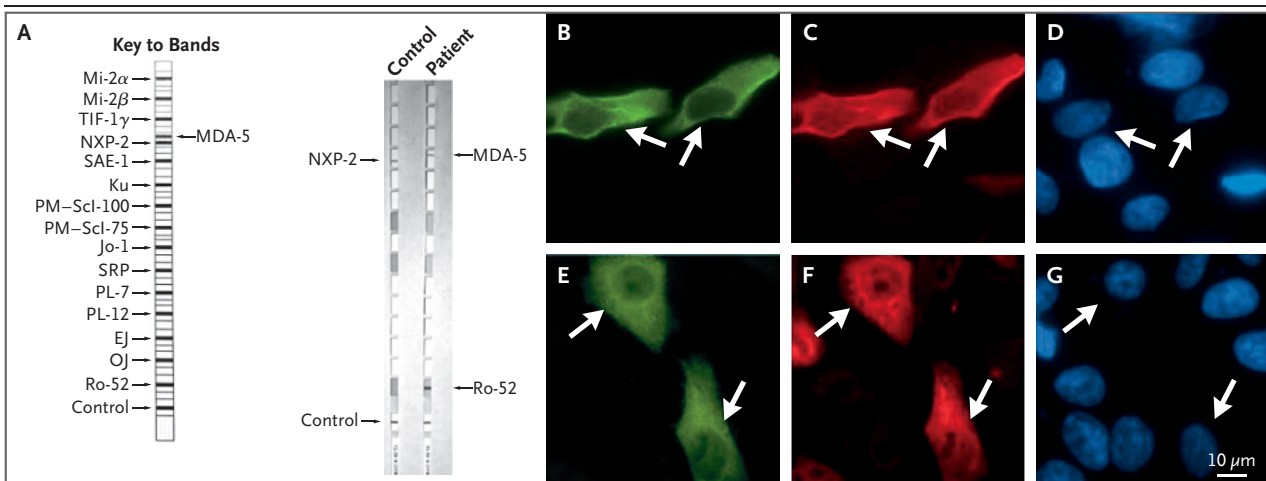


Figure 4. Test Results for Myositis-Associated Autoantibodies.

Panel A shows the results of an enzyme-linked immunosorbent assay used to detect myositis-associated autoantibodies. The assay consists of a test strip with 16 purified autoantigens arrayed in parallel bands. The 17th band is a positive control (human immunoglobulin) for the colorimetric reaction. After incubation with diluted patient serum, antibodies that were adherent to the test strip were detected with the use of enzyme-labeled anti-human IgG antibody. The positive control serum contained antibodies directed against NXP-2. The patient serum contained antibodies that reacted with a band that was slightly higher than the NXP-2 band, which corresponded to the location of MDA-5. The patient serum also contained antibodies that reacted with Ro-52.

Panels B through G show the results of an indirect immunofluorescence assay used to confirm the presence of anti-MDA-5 and anti-Ro-52 antibodies in the patient serum. To increase the levels of these autoantigens in HEp-2 cell substrate, HEp-2 cells were transiently transfected by plasmids encoding green fluorescent protein (GFP) fused to MDA-5 (Panels B, C, and D) or Ro-52 (Panels E, F, and G). Monoclonal anti-GFP antibody was used to identify successfully transfected cells (Panels B and E, in green). Antibodies in the patient serum stained only the cytoplasm of cells that expressed GFP-MDA-5 or GFP-Ro-52 proteins (Panels C and F, in red), a finding that confirmed the presence of autoantibodies. The patient serum did not contain antibodies directed against GFP alone (not shown). Staining with 4',6-diamidino-2-phenylindole dihydrochloride (DAPI) was used to show the location of nuclei (Panels D and G, in blue) in all cells, including those that did not express GFP-MDA-5 or GFP-Ro-52 proteins. The arrows indicate cells that expressed GFP fusion proteins.

positive results associated with some commercially available assays that are used to detect myositis-associated autoantibodies.¹⁶ Therefore, indirect immunofluorescence and HEp-2 cell substrate were used to confirm the presence of autoantibodies against MDA-5 and Ro-52 in the patient's serum. HEp-2 cells do not express MDA-5 and express very low levels of Ro-52, which explains the low titer of ANA in this patient. To increase the levels of these autoantigens in the HEp-2 cell substrate, HEp-2 cells were transiently transfected by plasmids encoding green fluorescent protein (GFP) fused to MDA-5 or Ro-52. The cells were subsequently fixed, permeabilized, and stained with the patient's serum. Antibodies in the patient's serum stained only the cells that expressed GFP-MDA-5 or GFP-Ro-52 (Fig. 4). Taken together, the results of the ELISA and the indirect immunofluorescence assay confirmed the presence of anti-MDA-5 and anti-Ro-52 antibodies in this patient.

PATHOLOGICAL DIAGNOSIS

Anti-melanoma differentiation-associated protein 5 (anti-MDA-5) dermatomyositis.

DISCUSSION OF MANAGEMENT

Dr. Jonathan Dau: We considered many treatment options for this patient, including high-dose glucocorticoid therapy, cyclophosphamide, rituximab, calcineurin inhibitors, intravenous immune globulin, plasmapheresis, and tofacitinib. Some patients with anti-MDA-5 dermatomyositis have rapidly progressive interstitial lung disease, and this patient had several risk factors for rapid progression, including the high ferritin level, presence of anti-Ro-52 antibodies, and Asian ethnic background.^{17,18} We took these factors into account when weighing the risk of death from rapidly progressive interstitial lung disease against the potential risk of opportu-

nistic infections associated with immunosuppression.

Limited data from case and historical control analyses showed that combination therapy with high-dose glucocorticoids, tacrolimus, and intravenous cyclophosphamide was associated with a 6-month survival rate of 89% in the treatment group, as compared with a rate of 33% in historical controls who received these therapies in a sequential addition.¹⁹ On the basis of these data, we initiated these three immunosuppressive drugs, in addition to intravenous immune globulin, on hospital day 14.

Despite this treatment, the patient's condition continued to worsen, and he required oxygen therapy through a high-flow nasal cannula. On hospital day 28, the patient began to receive mechanical ventilation. Treatment with tofacitinib, an oral inhibitor of Janus kinase 1 and 3, was initiated as salvage therapy on hospital day

34.²⁰ However, the next day, renal failure occurred. Hemodialysis with venovenous extracorporeal membrane oxygenation as a bridge to lung transplantation was considered. However, because of his smoking history, the patient was not considered to be a candidate for lung transplantation. On hospital day 37, after additional discussions with the patient's family regarding his poor prognosis, the goals of care were transitioned to comfort measures only, and the patient died the same day.

FINAL DIAGNOSIS

Anti-melanoma differentiation-associated protein 5 (anti-MDA-5) dermatomyositis.

This case was presented at the Medical Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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